

KOLBY JAYNIELLE LUIZ P. HERNANDEZ

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EDUCATION

Bachelor of Science, Information Technology – Cybersecurity 2023 - 2026
Mapua University – Makati Branch | GWA: 1.6923 | Cum Laude | DOST-SEI Scholar (RA 7687)

CERTIFICATIONS

- Google Cybersecurity**
Google | Issued Jun. 2025
Credential ID 1VRX3BGDO6H8
- Introduction to Cybersecurity Awareness**
HP LIFE | Issued Mar. 2026
Credential ID 30ff0262-6775-4d76-896c-c81b956f5dc3
- Certified Ethical Hacking (v12)**
LearnKartS | Issued Oct. 2025
Credential ID KMT5JGSHYI20

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL, C#, C++, PHP, JavaScript
Web Development: HTML5, CSS3, TypeScript, Wix, React, Next.js, Tailwind CSS, Framer Motion
Databases & Backend: MySQL, PostgreSQL, Supabase, Firebase, Node.js
Tools & Platforms: Git/GitHub, npm, Figma, Playwright, VS Code, Arduino IDE, Canva
Networking & Security Tools: Cisco Packet Tracer, Nessus, Kali Linux

EXPERIENCE

ADVENTUS IT SERVICES (PHILIPPINES) INC. Dec. 2025 - Apr. 2026
IT Support Trainee (Intern) Makati City, Philippines

- Conducted comprehensive User Acceptance Testing (UAT) on laptops, monitors, keyboards, mice, and headsets, systematically evaluating each unit's functionality and performance to confirm it met company quality standards and would satisfy end-user needs.
- Diagnosed and resolved hardware malfunctions and technical issues across multiple device types, ensuring equipment met reliability and performance benchmarks prior to deployment to clients
- Coordinated closely with the logistics team throughout the delivery process, verifying shipment accuracy and ensuring hardware reached the correct client on schedule.
- Maintained organized warehousing and storage protocols for defective hardware, properly logging and securing units pending repair or disposal to preserve inventory accuracy.
- Recognized as Best Intern of the Batch, an award reflecting consistent performance, reliability, and dedication throughout the 500-hour internship.

RAVA Digital Marketing Services August. 2025 – Dec. 2025
Junior Web Developer (Remote) Marikina City, Philippines

- Independently designed and developed a full e-commerce website on Wix for a wellness and therapy products brand, featuring a multi-category product catalog spanning physical devices (light therapy panels, terahertz wands, cold lasers), wearables, and digital healing content (guided audio/video courses)
- Delivered the completed website to a small wellness business based in Florida, USA, managing the project independently from planning through launch.

ACADEMIC PROJECTS

AMSIRS — Attendance Monitoring, Student Incident Reporting, Student Support

Technologies Used: Next.js 16, React 19, TypeScript, Supabase, Tailwind CSS, Framer Motion, Chart.js, Playwright, face-api.js

- Designed the front-end UI and architected the back-end and database schema for a full-stack campus system featuring facial recognition entry/exit gates with liveness detection and encrypted incident reporting.
- Implemented Row Level Security policies in Supabase/PostgreSQL and developed an AES-256-GCM encryption layer (Node.js crypto) to secure incident data at rest.
- Conducted a pre-launch security assessment covering API rate limiting, XSS testing, liveness-check spoofing attempts, and access control validation.

e-Barangay: Secure Document Request System

Technologies Used: HTML5, CSS3, JavaScript, Firebase, npm

- Served as Developer and Project Lead on a four-person team, building a digital service platform for a Barangay in the NCR area that modernizes and secures the barangay document request process through two-factor authentication, encrypted resident data, and QR-based document verification.
- Designed the interface in Figma and implemented the approved designs in HTML, CSS, and JavaScript.
- Built out the responsive behavior of the interface to ensure consistent usability across desktop and mobile devices.
- Implemented data encryption on the backend to protect resident information from unauthorized access and breaches.
- Integrated Firebase for user authentication and two-factor verification, along with QR-based validation to confirm the authenticity of official barangay documents.

SEMS — School Enrollment Management System

Technologies Used: PHP, MySQL, HTML, CSS, XAMPP (Apache)

- Developed a web-based enrollment management system for schools, building the full front end, back-end PHP logic, and MySQL database from the ground up to streamline student registration and administrative record-keeping.
- Built user authentication and role-based access control distinguishing admin and student/staff accounts, including registration, login, password management, and profile pages.
- Designed the relational database schema and integrated it with the PHP back end for CRUD operations on student and enrollment records.
- Structured the system for local deployment via XAMPP (Apache, MySQL, PHP, phpMyAdmin), including database setup and configuration documentation for future maintainers.

ACADEMIC PAPERS

Assessing CVE-2025-6218 or the WinRAR Path Traversal Exploit: Vulnerability Analysis and Simulation

- Co-authored a vulnerability analysis of CVE-2025-6218, a WinRAR directory-traversal flaw, reproducing the exploit with benign proof-of-concept files in an isolated virtual sandbox lab; performed root-cause analysis, compared vulnerable vs. patched builds, and developed mitigation recommendations including patch management, sandboxed extraction, and endpoint monitoring.
- Designed a controlled experiment across repeated trials that quantified a 100% exploit success rate for placing files into the per-user Startup folder under default WinRAR extraction settings.
- Received the highest grade in the course among submitted papers, recognized for both the research quality and the team's presentation.

Log4Shell Vulnerability: A Critical Analysis and Its Impact on Cybersecurity

- Co-authored a critical analysis of the CVE-2021-44228 or Log4Shell vulnerability, covering its exploitation mechanisms via JNDI/LDAP lookups, real-world impact on affected systems, and proposed honeypot and machine-learning-based defense mechanisms.
- Explained the technical root cause: Log4j's undocumented string-substitution feature let attackers embed a string in logged input, triggering a remote JNDI lookup that fetched and executed malicious Java code.

AlertoPH: Centralized Emergency Service Application

- Co-authored a study on a centralized mobile emergency-response platform for the Philippines featuring real-time GPS tracking, one-tap emergency dialing to police/fire/medical services, and community incident reporting.
- Served as the main designer of the application's user interface in Figma, building out the visual design and interactive proof-of-concept used to shape the app's core screens (login/sign-up, homepage, real-time map, hotlines).